

KAIZHEN TAN

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🏠 Personal Website

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RESEARCH INTERESTS

Focus: Urban Science × Embodied Intelligence

Topics: Robotic Urbanization, Human–Robot Interaction, Embodied AI (Spatial & Physical Intelligence)

Methods: Geospatial & Spatiotemporal Analysis, Computational Social Science, Causal Inference, Agent-Based Simulation, Digital Twins, Behavioral Modeling, Multimodal Perception, Representation Learning, World Models

EDUCATION

New York University

Ph.D. student in Urban Science

New York/Shanghai

2026 — 2031 (expected)

· Advisor: Prof. Chenghe Guan; Co-supervised by Prof. Zhan Guo

Carnegie Mellon University

M.S. in Artificial Intelligence

Pittsburgh

2025 — 2026

· GPA: 4.0/4.0; Courses: AI Model Development, LLM Systems, Agentic Technologies, AI Cloud and DevOps

Tongji University

B.Mgt. in Information Systems

Shanghai

2021 — 2025

· GPA: 3.94/4.0; Courses: System Design, Database Application, Deep Learning, Python/C++/Java

PUBLICATIONS

Journal Papers

[1] Huan, W., **Tan, K.**, Liu, X., Jia, S., Lu, S., Zhang, J., & Huang, W. (2025). A spatiotemporal adaptive local search method for tracking congestion propagation in dynamic networks. *GIScience & Remote Sensing*, 62(1), 2602215. DOI: 10.1080/15481603.2025.2602215

Conference Papers

[1] **Tan, K.** (2026). GeoFidelity-Bench: Evaluating Block-Conditioned Geographic Fidelity in Text-to-Image Street-View Generation. *Advances in Neural Information Processing Systems 39 (NeurIPS 2026)*. arXiv: 2606.23669

[2] **Tan, K.**, & Zhang, F. (2026). UrbanVGGT: Scalable Sidewalk Width Estimation from Street View Images. *The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences, XLIX-B2-2026* (XXV ISPRS Congress 2026), 559–567. DOI: 10.5194/isprs-archives-XLIX-B2-2026-559-2026

[3] **Tan, K.**, Feng, Y., & Du, H. (2026). CREG: Compass Relational Evidence Graph for Characterizing Directional Structure in VLM Spatial-Reasoning Attribution. *9th Chinese Conference on Pattern Recognition and Computer Vision (PRCV 2026)*. arXiv: 2603.20475

[4] **Tan, K.**, Gu, R., & Li, M. (2026). CacheWeaver: Cache-Aware Evidence Ordering for Efficient Grounded RAG Inference. *GroundLM @ EMNLP 2026*. arXiv: 2606.19667

[5] **Tan, K.**, Wu, Y., Liu, Y., & Zeng, H. (2025). Decoding Tourist Perception in Historic Urban Quarters with Multimodal Social Media Data: An AI-Based Framework and Evidence from Shanghai. *Global Smart Cities Summit cum The 4th International Conference on Urban Informatics*. arXiv: 2509.03830

[6] **Tan, K.** (2025). Multimodal Deep Learning for ATCO Command Lifecycle Modeling and Workload Prediction. *7th Asia Conference on Machine Learning and Computing (ACMLC 2025)*. DOI: 10.1145/3772673.3772702

Under Review & Preprints

[1] **Tan, K.**, Zheng, D., Wu, G., & Guan, C. (2026). PopNavShift: Stress-Testing Social Navigation under Behavioral Population Shift. *Under Review at IEEE International Conference on Robotics and Automation (ICRA 2027)*. arXiv: 2609.21838

[2] **Tan, K.**, Wu, G., Zheng, D., & Guan, C. (2026). Failure Has a Footprint: Rethinking Sidewalk Robot Failure in Public Space. *Under Review at ACM/IEEE International Conference on Human-Robot Interaction (HRI 2027)*.

- [3] **Tan, K.** (2026). From Robot Actions to Human Responses: What Can Public HRI Data Support? *Under Review at ACM/IEEE International Conference on Human-Robot Interaction (HRI 2027)*.
- [4] **Tan, K.**, Xu, S., Xu, X., Tao, S., Li, Y., Hong, H., Feng, Y., Du, H., & Wang, Z. (2026). What Can Latent World Models Know? Physical Parameter Identifiability in Multimodal Predictive Representations. *Under Review at ICLR 2027*. arXiv: 2607.27017
- [5] **Tan, K.**, Xu, X., Tao, S., Li, Y., Hong, H., Feng, Y., & Du, H. (2026). Beyond Multimodal Alignment: Certifying Physical Language through Response Substitution and Ordered Execution. *Under Review at ICLR 2027*. arXiv: 2608.19492
- [6] Deng, Y., Liu, J., **Tan, K.**, & Liu, Y. (2026). Early Learning Shapes Later Directions Of Representation Change In Continual Learning. *Under Review at ICLR 2027*. arXiv: 2609.36081
- [7] **Tan, K.** (2026). CityFresh: Temporal Validity as a Memory Layer for Urban Digital Twins. *Under Review for Future Cities in the Era of AI (CUPUM 2027)*.
- [8] **Tan, K.** (2026). Seeing the City or Recognizing the Place? What Street-View Imagery Adds Beyond Existing Urban Data in VLM Urban Sensing. *Preprint*. arXiv: 2610.00031
- [9] **Tan, K.** (2026). You Cannot Photograph the Same Street Twice: Reliability Limits in Vision–Language Measurement of Urban Change. *Preprint*. arXiv: 2609.00649
- [10] **Tan, K.**, & Deng, Y. (2026). Vision Models Predict Urban Scene Appraisal with Limited Neural Alignment. *Preprint*. arXiv: 2608.30964
- [11] **Tan, K.** (2026). Institutional Workflows and the Observed Geography of Metro Lost-Property Records: Evidence from Qingdao. *SSRN Preprint*. DOI: 10.2139/ssrn.7382978
- [12] **Tan, K.**, Feng, Y., Du, H., Tao, S., Xu, X., & Hong, H. (2026). Teaching Vision–Language Models to Use the Scale They Are Given: Label-Free Equivariance Training for Metric Physical Reasoning. *Preprint*. arXiv: 2609.00658
- [13] **Tan, K.**, Feng, Y., Du, H., Hong, H., Tao, S., & Xu, X. (2026). When Does Visual Token Pruning Improve Calibration? The Role of Evidence Coverage in MLLMs. *Preprint*. arXiv: 2604.12035

Working Papers

- [1] **Tan, K.** et al. (2026). Toward Human-Centered Robotic Urbanization: A Review for Technological, Spatial, and Institutional Co-evolution. *Working paper*.
- [2] **Tan, K.** et al. (2026). Robotic Urbanization and the Transformation of Ground Transportation Systems: Impacts, Challenges, and Opportunities. *Working paper*.
- [3] **Tan, K.** et al. (2026). RoboROW: Simulating and Governing Right-of-Way for Urban Service Robots in Shared Urban Spaces. *Working paper*.
- [4] **Tan, K.** et al. (2026). Legal Classification and the Morphological Selectivity of Delivery Robot Deployment: Comparing China and the United States. *Working paper*.
- [5] **Tan, K.** et al. (2026). Can Stair-Climbing Robots Improve Vertical Accessibility in Ageing Neighbourhoods? A Meal-Delivery Scenario Analysis in New York City. *Working paper*.

RESEARCH EXPERIENCE

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|---|--------------------------|
| <p>Shanghai Key Laboratory of Urban Design and Urban Science (LOUD)</p> <ul style="list-style-type: none"> · Advisor: Prof. <u>Chenghe Guan</u>, Prof. <u>Zhaonan Wang</u> · Study how embodied AI systems interact with urban space, infrastructure, human behavior, and governance, with a focus on robotic urbanization and human–robot interaction. | <p>05/2026 — Present</p> |
| <p>Peking University, Spatio-Temporal Social Sensing Lab (S3-Lab)</p> <ul style="list-style-type: none"> · Advisor: Prof. <u>Yu Liu</u>, Prof. <u>Fan Zhang</u> · Automated urban element measurement from street view using VGGT and semantic segmentation. · Recovered metric scale via ground plane fitting and camera height calibration for large-scale data generation. | <p>10/2025 — 03/2026</p> |

A*STAR, Institute for Infocomm Research, Singapore

09/2024 — 05/2025

- Advisor: Dr. Yicheng Zhang, Dr. Sheng Zhang
- Modeled air traffic controller communication tasks to predict workload using radiotelephony and trajectory data.
- Integrated multi-source air traffic data and evaluated scenarios with a CNN-Transformer model.

Tongji University, College of Architecture and Urban Planning

04/2024 — 04/2025

- Advisor: Prof. Yujia Zhai
- Built a multimodal pipeline with social media reviews and photos to analyze tourist perception of historic quarters.
- Fine-tuned segmentation models and applied sentiment analysis to produce multi-dimension satisfaction scores.

Tongji University, College of Surveying and Geo-Informatics

04/2024 — 12/2024

- Advisor: Prof. Wei Huang
- Modeled traffic congestion propagation with spatiotemporal graphs and multi-scale community search.
- Linked propagation patterns with built environment factors using causal analysis and POI-based features.

ACADEMIC PROJECTS

Urban Vision-Language Guide Agent for Visually Impaired People

10/2025 — 12/2025

- Design a real-time egocentric navigation assistant that provides spoken guidance to blind users while walking.
- Build a lightweight perception and event engine to segment key urban classes, estimate depth, and convert detections into concise voice prompts (e.g., “enter tactile paving,” “obstacle left/right,” “traffic light walk/stop”).

Pavement Disease Recognition for Smart Road Maintenance

09/2023 — 11/2023

- Enhanced YOLOv8 for small-target recognition and applied Wise-IoU, improving F1 by 26.3%.
- Applied AI to urban infrastructure monitoring, winning Second Prize at the national AI competition (top 5%).

Agent-based Modeling and Simulation for Public Facility Optimization

06/2023 — 08/2023

- Collected behavioral data on library seat usage and applied clustering & FAHP analysis.
- Built an AnyLogic-based simulation system for crowd behavior modeling, achieving 0.87 grey correlation.

INDUSTRY EXPERIENCE

Shanghai Artificial Intelligence Lab, AI Product Manager Intern

01/2025 — 04/2025

- Conducted end-to-end market research and competitive analysis for AI products (e.g., Migo, Auto-Research), benchmarking against 50+ vertical tools and delivered 5+ analytical reports to guide product positioning.
- Built operational data dashboard tracking KPIs, enabling granular analysis of user retention and behavior funnels.
- Established evaluation frameworks for Migo’s InternLM model, enabling data-driven iteration cycles.

ACADEMIC SERVICE

Conference Reviewer: NeurIPS 2026; AAAI 2027

SKILLS

Languages: IELTS 7.5 (L:9, R:8.5, W:6.5, S:6), GRE 331 (V:161, Q:170, AW:3)

Programming & Tools: Python, C++, C#, Java, HTML, SQL, Git, Docker, Tableau, ArcGIS, Stata